

Worked probs

* Ran Colab on CPU $\therefore$ the old config's TPU no longer available on Colab

i) Ops:

• Matmul: %dot.2

Output: f32[4096]

Input: f32[8192], f32[4096, 8192]

rhs contracting dim = {1} $\Rightarrow$ matrix's second dim contracts $\rightarrow$ vector dim

fw, bw $\rightarrow$ bf

• Matmul: %dot.3

Output = f32[8192]

Input = f32[4096], f32[8192, 4096]

wf, bf $\rightarrow$ bw

• AllReduce: %all-reduce.1

Output: f32[1, 8192] in row-major order.

[1, 8] $\leq$ [8] $\Rightarrow$ one group contains all 8 devices.

$\therefore$ Total dim since each device owns 4096 of global FFW dim

$\Rightarrow$ total dim = $4096 \times 8 = 32768$

$\therefore$ output shape final = [1, 8192]

$\therefore$ What's going on here: FFW layer sharded into 8 devices, then the partial ~~outputs~~ outputs of each device's matmul is then AllReduced to get the final output of shape 8192.

What may look suspicious: The final AllReduce communicating full [1, 8192] output across all 8 devices. But it is necessary because for the 2nd matmul, the dimension that is contracted is also the one that is sharded $\Rightarrow$ $\therefore$ Each device only has a partial result $\Rightarrow$ need AllReduce $\Rightarrow$ 8-way model parallelism